

Part 4: Fragmentation Functions

The Non-Perturbative Object That Makes Factorization Work

PHYS 7502 – Final Reading Assignment

1 The Problem Fragmentation Functions Solve

You’ve already lived through this idea on the “incoming” side: when computing a DIS cross section, you can’t use a proton directly in a Feynman diagram. Instead, you factorize:

$$\sigma(ep \rightarrow e + X) = \int dx f_{q/p}(x, Q^2) \hat{\sigma}(eq \rightarrow eq) \quad (1)$$

where $f_{q/p}(x, Q^2)$ is the **parton distribution function** (PDF) – the probability of finding a quark inside the proton carrying momentum fraction x . The PDF is non-perturbative (we can’t compute it from QCD first principles), but the partonic cross section $\hat{\sigma}$ is perturbative.

What’s Symmetrical Here?

The same logic should work on the *outgoing* side. If you make a quark in an e^+e^- collision, that quark doesn’t stay a quark – it becomes a spray of hadrons. To compute the cross section for producing a specific hadron h in the final state, you want something like:

$$\sigma(e^+e^- \rightarrow h + X) = \hat{\sigma}(e^+e^- \rightarrow q\bar{q}) \otimes (\text{probability that } q \rightarrow h \text{ with some momentum fraction})$$

That “probability that $q \rightarrow h$ ” is the **fragmentation function**.

So a fragmentation function is just the “outgoing” analog of a PDF. In the old literature they were sometimes called *parton decay functions*, which actually expresses the idea quite clearly.

2 Definition and Notation

Fragmentation Function

The fragmentation function $D_i^h(z)$ is the probability density that a parton of type i (quark, antiquark, or gluon) “fragments” into a hadron of type h carrying momentum fraction z of the parton’s momentum:

$$D_i^h(z) dz = \text{number of hadrons } h \text{ inside parton } i \text{ with longitudinal momentum fraction in } [z, z+dz] \quad (2)$$

Here “longitudinal” means along the direction of the parton’s motion. The transverse momentum of the hadron relative to the parton is integrated over in this definition (we’ll call this variant the *integrated* or *collinear* FF below).

2.1 The notation mess: x versus z

This is an important one. *Both* PDFs and FFs use a momentum fraction variable, and historically different people have used different letters. Here's the deal:

Convention Clash: x vs. z

In PDFs (the incoming side): x = fraction of the *incoming hadron's* momentum carried by the parton that actually scatters.

In FFs (the outgoing side): z (or sometimes also x !) = fraction of the *outgoing parton's* momentum carried by a specific hadron.

These are *completely different* quantities despite the overlap in notation. In Griffiths' DIS treatment (Edition 1), the symbol x is used for the first thing. If you read a paper on $e^+e^- \rightarrow$ hadrons that uses x , it probably means the second thing.

In this course: we'll use z for the fragmentation variable to keep the distinction clear. Most modern literature does the same.

2.2 Variants of FFs

Just like there's no single PDF (there are u -quark PDFs, d -quark PDFs, gluon PDFs, polarized PDFs, etc.), there's no single FF. Different physical quantities you want to predict require different variants:

- **Integrated (collinear) FF** $D_1^{h/i}(z)$: the simplest version, integrated over the hadron's transverse momentum relative to the parton. This is the one that appears in inclusive cross sections. The subscript 1 is a convention meaning leading-twist.
- **Transverse-momentum dependent (TMD) FF** $D^{h/i}(z, \vec{P}_{hT})$: keeps the dependence on the hadron's transverse momentum $\vec{P}_{hT}$. Needed when you measure things differential in transverse kinematics.
- **Polarized FFs**: if the parent parton has spin, or the produced hadron has spin, you get additional spin-dependent FFs like $G_1^{h/i}$, $H_1^{h/i}$. The *Collins function* $H_1^{\perp h/q}$ is one such TMD-FF that's played a major role in spin physics.
- **Higher-twist and di-hadron FFs**: for sub-leading-power effects or when you identify pairs of hadrons in the final state.

In this reading we focus on the simplest case: $D_1^{h/i}(z)$, the integrated unpolarized FF.

3 Where FFs Appear: Three Key Processes

FFs appear in the factorization theorems for any process that identifies a specific final-state hadron. The three workhorses are:

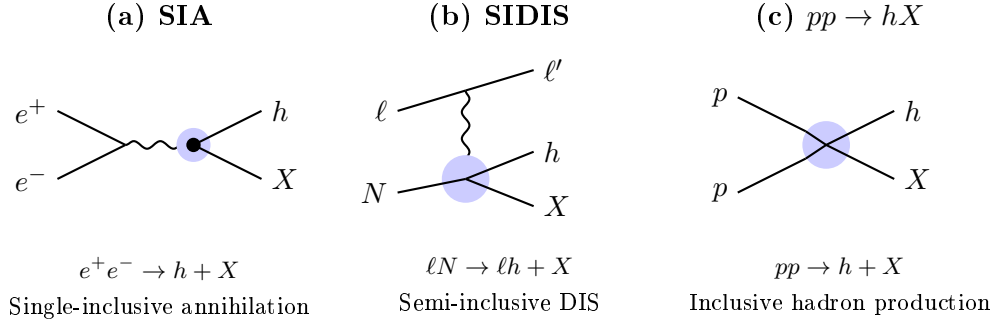


Figure 1: The three main processes where fragmentation functions appear. In all cases, an identified hadron h is produced alongside unspecified additional activity X . The “blob” represents the non-perturbative region where FFs live.

The corresponding schematic factorization theorems are:

$$\sigma(e^+e^- \rightarrow h + X) = \hat{\sigma} \otimes FF \quad (3)$$

$$\sigma(\ell N \rightarrow \ell h + X) = \hat{\sigma} \otimes PDF \otimes FF \quad (4)$$

$$\sigma(pp \rightarrow h + X) = \hat{\sigma} \otimes PDF \otimes PDF \otimes FF \quad (5)$$

Reading These Equations

Each “ $\otimes$ ” represents a convolution integral over momentum fractions. The pattern is:

- **SIA** ($e^+e^- \rightarrow h + X$): No incoming hadrons, so no PDFs – just the hard parton cross section times an FF. This is the *cleanest* place to measure FFs.
- **SIDIS**: A lepton beam on a nucleon target. Need one PDF (for the parton in the nucleon) and one FF (for the hadron coming out of the struck parton).
- $pp \rightarrow hX$: Two incoming hadrons $\rightarrow$ two PDFs, and one outgoing identified hadron $\rightarrow$ one FF.

3.1 What makes factorization work?

Factorization theorems hold only under specific kinematic conditions, the minimum being the presence of a **hard scale** Q so that perturbative QCD can be used:

- For SIA: $Q = \sqrt{s}$ (the center-of-mass energy).
- For SIDIS: Q = the momentum transfer from the lepton to the hadronic system (the Q^2 you know from DIS).
- For $pp \rightarrow hX$: $Q = p_T$ of the final-state hadron relative to the beam axis.

Factorization holds as an expansion in $1/Q$, and the equations above are the *leading* term. Subleading terms exist but are much harder to prove rigorously.

4 The Webber Factorization Formula

Now let's get more quantitative. The main factorization formula for single-inclusive annihilation $e^+e^- \rightarrow h + X$ (from Webber) is:

Webber's Master Equation

$$F^h(x, s) = \sum_i \int_x^1 \frac{dz}{z} C_i(z, \alpha_s(s)) D_i^h(x/z, s) \quad (6)$$

where:

- $F^h(x, s)$ is the measured inclusive hadron distribution.
- $x = 2E_h/E_{\text{cm}}$ is the scaled hadron energy.
- $s = q^2$ is the center-of-mass energy squared.
- $C_i(z, \alpha_s)$ is the **coefficient function** for the hard process producing parton i , including all the process-specific selection cuts.
- $D_i^h(\xi, s)$ is the **fragmentation function** for parton $i \rightarrow h$.
- The sum runs over all parton flavors i .

Take a moment to parse this. At leading order, the coefficient function C_i is just proportional to a delta function $\delta(1 - z)$ – the parton carries all the CM energy, no radiation. Then the formula becomes simply $F^h(x, s) = \sum_i \hat{\sigma}_i D_i^h(x, s)$, which is the naive parton-model result.

Higher-order corrections make C_i nontrivial (it gets $\log z$ terms from soft/collinear emissions, etc.), but the basic structure – hard cross section convolved with FF – stays the same.

5 Scaling Violations: DGLAP for FFs

Just like PDFs, fragmentation functions depend on the energy scale $Q^2 = s$. This is called **scaling violation** (if there were no Q^2 dependence, we'd call it “scaling”, and Bjorken showed that's the naive parton-model prediction).

The scale dependence is governed by an equation of exactly the same form as the DGLAP equation for PDFs:

DGLAP Evolution for FFs

$$s \frac{\partial}{\partial s} D_i^h(x, s) = \sum_j \int_x^1 \frac{dz}{z} P_{ji}(z, \alpha_s(s)) D_j^h(x/z, s) \quad (7)$$

where $P_{ji}(z)$ are the **splitting functions** – the same ones that appear in the PDF DGLAP evolution, but with a subtle difference: the indices are “transposed” between the two cases (for FFs we're asking about $j \rightarrow i$ rather than $i \rightarrow j$).

Why Scaling Violation?

Imagine a quark at scale Q , about to fragment into a hadron h at fraction z . If we go to *higher* Q , that quark can first emit a (harder) gluon – so its momentum gets split, and the hadron it eventually produces ends up carrying a *smaller* fraction of the original. Higher $Q \rightarrow$ more low- z hadrons, fewer high- z ones. This is exactly what DGLAP evolution predicts.

Practical consequence: we *parameterize* the FFs at some low reference scale s_0 (e.g., $s_0 = m_Z^2$), fit the parameters to data, and use equation (7) to *evolve* to other energies.

6 Small- x Physics and the Hump-Backed Plateau

Here's a phenomenon that's characteristic of the soft end of jet spectra.

6.1 The problem with small x

At very small x (very soft hadrons), the DGLAP evolution picks up *large logarithms* of x . Specifically, each order in α_s contributes terms like $\alpha_s \log(1/x)$. When x is small enough, these logs compensate for the smallness of α_s , and the fixed-order expansion breaks down.

6.2 MLLA: the modified leading log approximation

The fix is to **resum** all those large logs. This is done by a modification of DGLAP called the **Modified Leading Log Approximation (MLLA)**:

$$s \frac{\partial}{\partial s} D_i^h(x, s) = \sum_j \int_x^1 \frac{dz}{z} P_{ji}(z, \alpha_s(s)) D_j^h(x/z, \mathbf{z}^2 \mathbf{s}) \quad (8)$$

Notice the z^2 in the argument of D_j^h on the right. This change captures the dominant effects of coherent soft gluon emission: emissions are *angle-ordered*, and the effective scale for the next emission is reduced by the momentum fraction z .

6.3 The hump-backed plateau

The characteristic experimental signature of MLLA is a **hump-backed** shape when you plot the hadron distribution against $\xi = \ln(1/x)$:

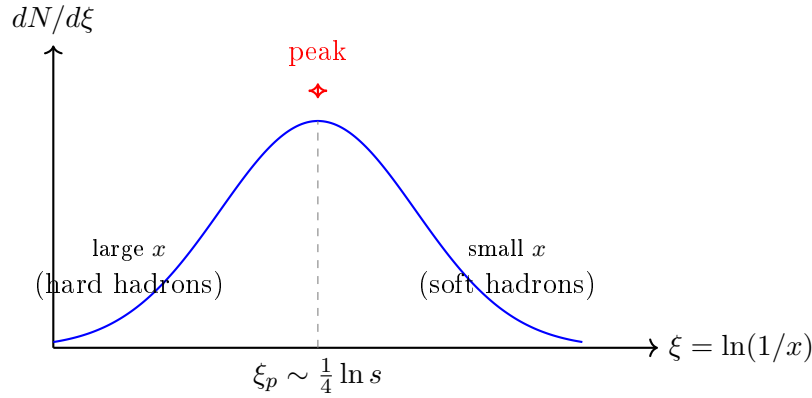


Figure 2: The hump-backed plateau. Plotting the number of hadrons per unit $\xi = \ln(1/x)$ as a function of ξ gives a characteristic hump whose peak position grows as $\xi_p \sim \frac{1}{4} \ln s$. The shape is a prediction of MLLA and agrees beautifully with data over a wide range of energies, from about $\sqrt{s} = 14 \text{ GeV}$ up to LEP and beyond.

What the Hump Tells Us

The hump is direct evidence that soft gluon *coherence* is real. If emissions were incoherent, you'd just get more and more soft hadrons as $\xi \rightarrow \infty$ (i.e., $x \rightarrow 0$) – no hump, just a plateau that grows forever. The fact that we see a hump with a calculable peak location is a striking confirmation of the QCD shower picture.

7 Factorization Scheme and Its Price

A subtle but important point. Although fragmentation functions are **universal** (same FF works in e^+e^- , DIS, pp), they are **factorization-scheme dependent**. The splitting functions P_{ji} depend on scheme beyond leading order.

What Does “Scheme” Mean Here?

When you go beyond leading order, there's an ambiguity: how much of the finite part of the NLO correction do you absorb into the FF, and how much into the coefficient function? The two choices give the same physical cross section, but different FFs and different C_i . Different conventions are called different “schemes” – the most common is $\overline{MS}$ (“MS-bar”).

Practical consequence: quoted FFs always come with a scheme label, and you can't mix and match (an FF fit in $\overline{MS}$ needs to be used with NLO coefficient functions also in $\overline{MS}$).

8 Quark versus Gluon Fragmentation: the Big Qualitative Result

The reading emphasizes this point: at the end of Webber's talk, he contrasts fragmentation functions for quark-initiated versus gluon-initiated jets. This is conceptually important and has real experimental implications.

8.1 The key physical difference

The gluon carries higher color charge than the quark. In QCD color factors:

$$C_A = 3 \quad (\text{gluon color charge}), \quad C_F = \frac{4}{3} \quad (\text{quark color charge}) \quad (9)$$

(You’ve calculated both of these in your homework – C_F came out of qq scattering via gluon exchange, C_A comes from the gluon self-interaction.)

Higher color charge means:

- More soft gluon radiation.
- More splittings during the shower.
- Therefore: **higher multiplicity** and **softer spectrum** in gluon jets compared to quark jets.

8.2 The multiplicity ratio

Take the ratio of charged-particle multiplicities in gluon jets vs. quark jets at low rapidity (large angle from the jet axis):

$$\frac{N_g(\text{ch})}{N_q(\text{ch})} \rightarrow \frac{C_A}{C_F} = \frac{9}{4} = 2.25 \quad (10)$$

in the asymptotic limit. In practice, OPAL measured

$$r_{\text{ch}}(p < 4 \text{ GeV}, 0.8 < p_T < 3 \text{ GeV}) = 2.29 \pm 0.09 \pm 0.015, \quad (11)$$

a beautiful confirmation of the QCD prediction. This single measurement directly verifies the color-charge difference between quarks and gluons in the hadronic final state.

Quark vs. Gluon Jets, Experimentally

Property	Quark jet	Gluon jet
Color charge (Casimir)	$C_F = 4/3$	$C_A = 3$
Multiplicity	Lower	Higher (by $\sim C_A/C_F$)
x spectrum	Harder	Softer
Rapidity profile	Narrower	Wider

8.3 How do we identify quark vs. gluon jets experimentally?

At a collider like LEP ($e^+e^- \rightarrow q\bar{q}g$ three-jet events), you can *anti-tag* the quark jets by identifying them as b-jets (via displaced vertices), leaving the third jet as the gluon jet. This was DELPHI’s and OPAL’s approach. Modern ATLAS/CMS also use machine-learning techniques on substructure variables.

8.4 No species-level enhancement

One striking null result: the relative particle species yields (pion:kaon:proton) are roughly the same in quark and gluon jets. The difference is in *how many total particles* there are, not in which species. This supports the idea of **local parton-hadron duality**: hadronization is a local process that doesn’t care much about whether its parton input was a quark or a gluon, as long as the total color flow is the same.

9 Connecting to the Rest of the Reading

Why do FFs matter for the rest of this hadronization story?

- The Lund string model (Part 6) *derives* a specific functional form for the FF:

$$f(z_+, m^2) = N z_+^{-1} (1 - z_+)^a e^{-bm^2/z_+}$$

the **Lund symmetric splitting function**. This is used inside PYTHIA (Part 3).

- The Dual Parton Model (Part 7) makes predictions for FFs in various channels (quark $\rightarrow$ pion, quark $\rightarrow$ baryon, diquark $\rightarrow$ baryon, etc.). Compare:

$$xD_{\pi/q}(x) = (1 - x), \quad xD_{b/q}(x) = x^\alpha(1 - x)^3$$

- Ultimately, factorization + FFs is how theorists turn QCD predictions into hadron-level cross sections that can actually be measured at colliders.

10 Summary of Part 4

Main Takeaways

1. A **fragmentation function** $D_i^h(z)$ is the probability density for a parton i to fragment into a hadron h carrying momentum fraction z . It's the outgoing counterpart of a PDF.
2. **Watch out for x vs. z .** In PDFs, x is the incoming parton momentum fraction; in FFs, z is the fraction of the outgoing parton's momentum carried by the hadron. Don't confuse them.
3. FFs appear in factorization for **single-inclusive annihilation (SIA)**, **semi-inclusive DIS (SIDIS)**, and **inclusive hadron production in pp** . SIA is the cleanest place to measure them.
4. FFs satisfy their own version of the **DGLAP evolution equation** – parameterize at one scale, evolve to others. This is called *scaling violation*.
5. At small x , large logs force resummation via the **MLLA**, producing the characteristic **hump-backed plateau** in $\xi = \ln(1/x)$ distributions.
6. FFs are **universal** (same FF in e^+e^- , DIS, pp) but **factorization-scheme dependent** beyond LO.
7. **Gluon jets have higher multiplicity than quark jets**, by a factor approaching $C_A/C_F = 9/4 \approx 2.25$. OPAL has measured this ratio consistent with the QCD prediction.